

# [DC] CoboDeck Pro: Advanced Encounter-Typed Haptic Device for Collaborative Architectural Design in Walkable VR

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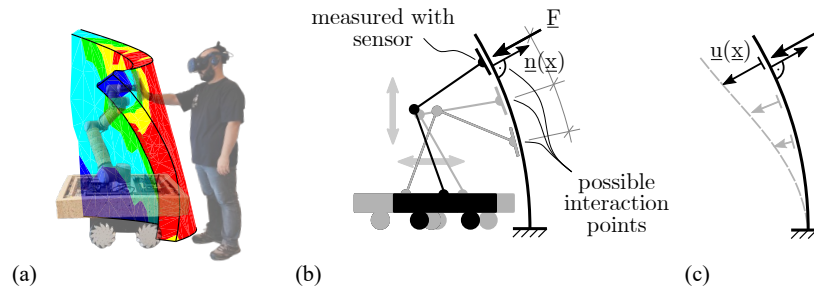


Figure 1: (a) A user interacting with a physical plate-prop presented by the robot in the real world, while simultaneously viewing the deformation of the virtual wall and corresponding stress fields in VR. (b) A simplified 2D schematic showing the user's applied force (c) The resulting displacement upon loading by the user is mimicked through the retracting robot arm.

## ABSTRACT

Haptic feedback has consistently shown its potential to enhance task performance in virtual environments. In architectural design, the ability to sketch and conceptualize directly in Virtual Reality (VR), rather than relying on traditional methods, offers a transformative shift in the design process. Integrating haptic feedback into this process can further improve design practices. This dissertation introduces CoboDeck Pro, an advanced haptic device designed to simulate structural deformation, enabling more informed decision-making during early design stages. Additionally, the research addresses multi-user haptic interactions to foster interdisciplinary collaboration while reducing system costs per user. The proposed platform integrates into architectural design frameworks, making the process more collaborative, interactive, and responsive.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices; Computer systems organization—Embedded and cyberphysical systems—Robotics;

## 1 BACKGROUND & MOTIVATION

The architectural, engineering, and construction (AEC) workflows are inherently complex and require interdisciplinary collaboration. Traditional processes are largely sequential, starting with paper sketches, progressing to digital refinements and visualizations, and addressing structural design only in later stages. Although structural considerations are crucial for material selection and ensuring buildings can safely resist loads, they are often overlooked in early design stages, where their impact is most significant.

Virtual Reality (VR) offers immense potential to overcome the limitations of traditional design workflows by enabling architects and structural engineers to collaborate more effectively [5]. Integrating haptic into VR sketching further enhances this process by allowing designers to gain hands-on experience, both seeing and feeling the mechanical responses of their designs with various com-

binations of geometry, structure, and material. Such immersive interactions make the early design stages more tangible and interactive, facilitating the exploration of structural behavior and material selection much earlier than is typical in standard workflows [3].

A specific class of haptic technologies, known as Encounter-Typed Haptic Devices (ETHDs) utilize robots to provide real-time haptic feedback, enabling users to touch and interact with virtual objects as if they were physically present [1]. ETHDs offer dynamic and versatile feedback, making them particularly suitable for arbitrary 3D sketches and architectural design. CoboDeck [2], an ongoing project at Vienna University of Technology, exemplifies ETHDs in a large-scale walkable VR setting. It features an omnidirectional mobile platform with an attached robotic arm that presents physical props at virtual touchpoints, matching their position and orientation. This allows users to intuitively interact with virtual objects using bare hands in a natural way without being aware of the robot's real-world operation.

While CoboDeck shows promise for supporting creative architectural design in VR, it has key limitations that restrict its applicability in structural analysis. First, it can only present simple, rigid props, lacking the ability to simulate diverse curvatures or mechanical behaviors. This is particularly critical in structural analysis, where feeling and visualizing the elastic response and identifying stress points of structures under force is essential. Second, the system is limited to single-user interactions, whereas mutual understanding and collaboration among interdisciplinary teams are vital for informed decision-making in AEC design. Additionally, the high cost of the robot and equipment for serving a single designer reduces its practicality. A multi-user setup could address this by allowing multiple designers to share the system, leveraging idle robot time (more than 50% of operation time) to serve additional users.

This research aims to introduce "CoboDeck Pro" (Figure 1) by addressing the limitations of current CoboDeck and improving its integration into AEC design frameworks.

## 2 PROPOSED RESEARCH

### 2.1 Haptic Simulation of Deformation

To simulate the deformation of virtual structures under applied forces, surface displacement and resistance based on the force exerted by the user and the mechanical properties of the material must be calculated. The robotic arm should then react by either resisting

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or retracting accordingly, providing users with an authentic sense of how the structure would behave under real-world conditions. Exploring solutions to create effective and realistic haptic feedback that aligns with the capabilities of the platform and user expectations is a central challenge of this component.

**RQ1.** How can the real-time deformation of virtual structures be simulated and conveyed through haptic feedback with minimum latency and maximum flexibility?

## 2.2 Multi-user support

To enable multi-user support, the robot behaviors must be re-designed. A key challenge in ETHDs is the delay in preparing and presenting haptic props. Addressing this, one research goal is to investigate acceptable levels of haptic delay for VR users through user studies targeting relevant scenarios.

Multi-user interaction introduces additional complexities, as the robot must dynamically allocate resources to serve multiple users simultaneously. This requires optimizing resource sharing, such as prioritizing users based on their immediate need for haptic interaction. One aspect of this optimization is determining the robot's position in the workspace. The robot must strategically position itself to minimize response times while avoiding collisions, ensuring a balance between user experience, performance, and safety.

**RQ2.** What levels of haptic delay are acceptable for VR users, and how do delays impact realism and usability?

**RQ3.** How can robot positioning be designed to efficiently serve multiple users while ensuring safety?

## 2.3 Integration into Design Frameworks

AEC workflows involve collaboration between architects, structural engineers, and material scientists. Integrating CoboDeck Pro into these workflows requires seamless interaction between the haptic platform and structural and material modeling tools. This involves real-time synchronization of deformation data to ensure accurate and responsive feedback at the interaction point. Achieving this requires a tightly integrated digital design framework, supported by interdisciplinary collaboration among researchers, to deliver early-stage, reliable feedback for designers.

**RQ4.** How can haptic feedback be effectively integrated with structural and material modeling to create a seamless digital design framework with real-time feedback for stakeholders?

## 3 RESEARCH PROGRESS

### 3.1 Haptic Simulation of Deformation Progress

To simulate deformation in CoboDeck Pro, a force-torque sensor integrated into the robot arm measures user-applied forces in real-time (Figure 1.b). The force  $F$  is then multiplied by a precomputed material-dependent coefficient to calculate the corresponding surface displacement (Figure 1.c). To ensure realistic and responsive haptic feedback, the robotic arm's prop displacement is implemented using closed-loop control of the arm joints, reducing reaction delays compared to probabilistic path planners. However, closed-loop control alone faces challenges in scenarios requiring high flexibility due to inherent delays and the limited flexibility of the arm hardware. To address this, we are exploring origami-inspired end-effectors. The first prototype, the Zipper Flower Tube [4], demonstrates how combining origami structures with minimal mechanical actuation creates a flexible, responsive, and cost-effective prop.

Future work will focus on searching for new origami designs and integrating them with closed-loop control methods. This hybrid approach aims to leverage the inherent responsiveness of the origami end-effector to compensate for the initial lag of the robotic arm.

### 3.2 Multi-user support Progress

The main challenge in multi-user support is minimizing the robot's response time. To address this, we plan to begin with a user study to evaluate how haptic delay impacts the VR experience and determine the maximum tolerable delay for our application.

Currently, the robot's positioning in the workspace in CoboDeck relies on a hand-crafted algorithm. To improve efficiency, we are developing a data-driven approach using Deep Reinforcement Learning (DRL) in the Nvidia Isaac Lab environment. The next goal is to predict user intentions, allowing the robot to position closer to the user most likely to request haptic feedback. If the DRL-based optimization for robot positioning shows promising results, the next step will involve extending it to arm navigation, making all robot movements integrated.

Despite these optimizations, scenarios where multiple users simultaneously request haptic feedback are inevitable. Resolving such conflicts will require strategies that minimize negative impacts on user immersion and task performance. Future research will focus on designing and evaluating these strategies to ensure efficient multi-user haptic interaction.

### 3.3 Integration into Design Frameworks Progress

In collaboration with architects, material scientists, and structural engineers, we developed a novel digital design framework that facilitates active participation from all stakeholders in design decisions [3]. The concept and benefits of incorporating haptic feedback into structural design have been demonstrated through two use cases described in the paper. As development on other components of CoboDeck Pro progresses, future work will focus on enhancing this framework, making it more robust and practical.

## 4 CONCLUSION

This research explores the integration of haptic feedback into architectural design workflows through the development of CoboDeck Pro, an enhanced large-scale ETHD. The project focuses on implementing haptic simulation of deformation to enable early structural analysis and scaling the platform to support multi-user interaction. Additionally, the research aims to seamlessly integrate CoboDeck Pro as a haptic module within architectural design frameworks. To conclude, the main areas I would like feedback on are: 1) Are there specific measures or experiments to consider for evaluating haptic delay in VR applications? 2) What questions or topics within my PhD would benefit the community most? 3) What approaches could be the base for managing simultaneous haptic requests effectively?

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